BABY

Build A Better Yield blanket

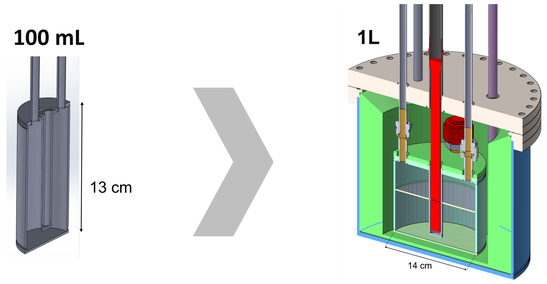
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MIT Plasma Science and Fusion Center

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Scaling up

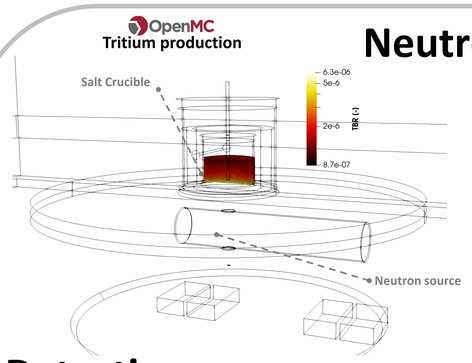


BABY specs

Salt volume	$0.1 - 1$ L
TBR	$\approx 10^{-3}$
Temperature	700 °C
DT Neutron energy	14 MeV

Neutrons

Neutronics



- Neutronics simulations track the path of neutrons
- Used to predict tritium generation, neutron flux, heating, production of activated materials…

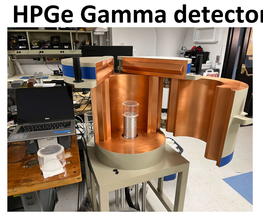
Detection

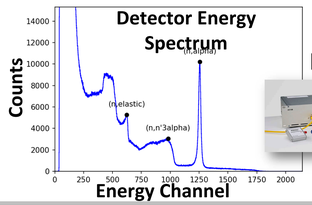
- LIBRA will provide neutronics data for validating codes
- Activation foils and diamond detectors as used to characterize the neutron sources and track the neutron flux

Activation foils



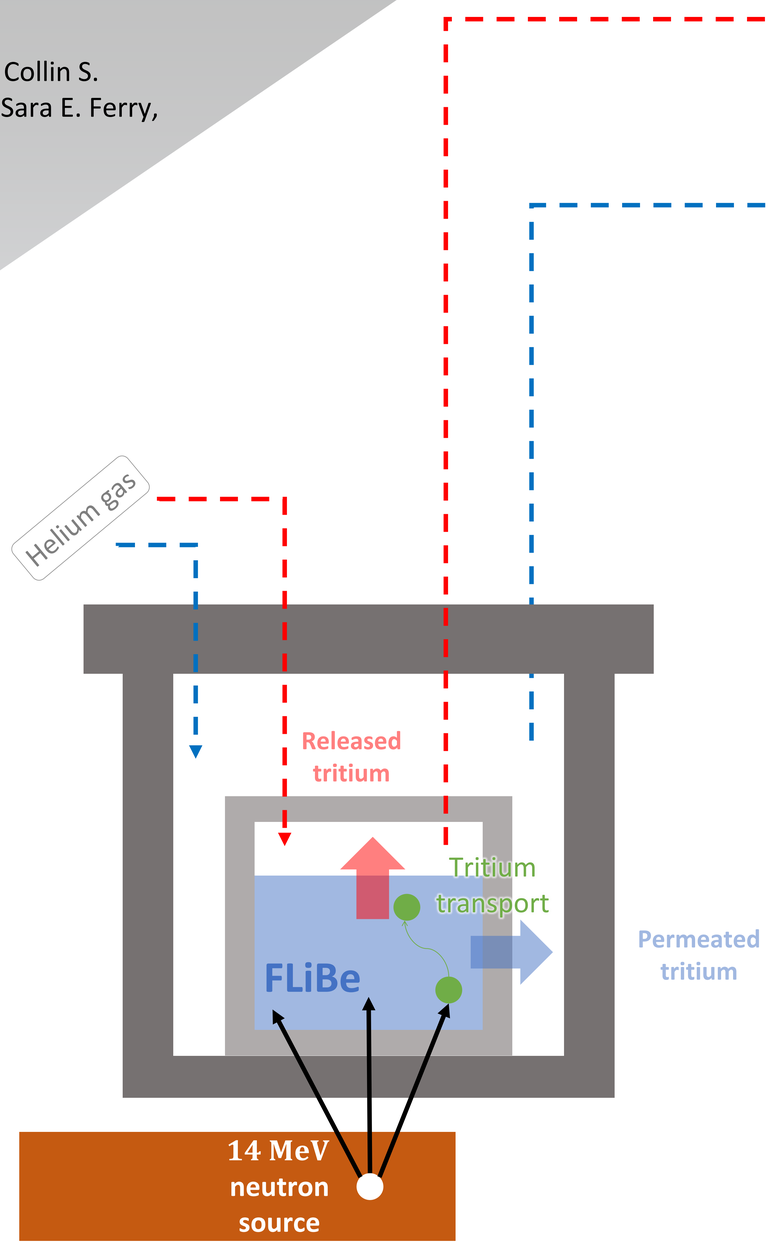
HPGe Gamma detector





Diamond detector



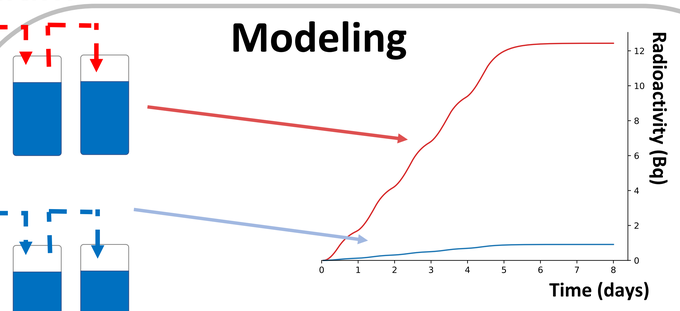


1L setup



Tritium

Modeling



0D tritium transport model:

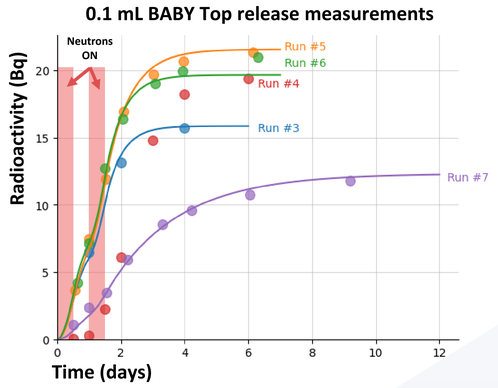
$$V\frac{\partial c}{\partial t} = S - A_{\mathrm{p}}\ k_{\mathrm{p}}\ c - A_{\mathrm{r}}\ k_{\mathrm{r}}\ c$$

V: salt volume
c: salt tritium concentration
S: tritium source
A_i: surface area
k_i: mass transport coefficient

Detection

Tritium water bubblers

- Bubblers collect 99% of the tritium
- Water samples are analyzed with Liquid Scintillation Counting
- Measurements on BABY serve as validation data

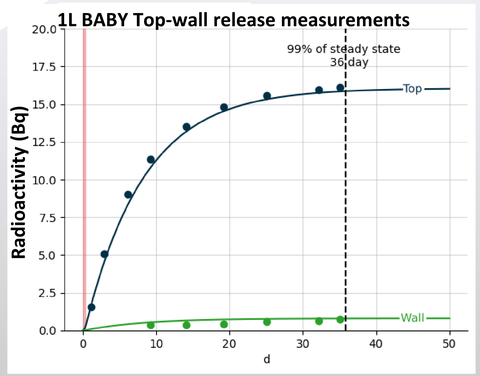


1ml results

- Bred tritium
- Validated neutronics and tritium transport models
- Good reproducibility
- Tested salt chemistry effect on tritium dynamics

1L baby specs

- Higher tritium breeding ratio
- Can collect wall-permeated tritium
- Implements tritium dynamics acceleration technologies
- Tests tritium permeation barriers
- Quantify and refine uncertainties



1L baby results

- Scaled up tritium breeding
- Validated neutronics and tritium transport models
- Informed tritium dynamics model parameters (volume, breeding rate)